

The Two Formulas for a Hit Romance Novel

A computational analysis of what separates market **reach** from reader **quality** in 92 billionaire romance novels.

Core Finding: We discovered that success in romance isn't a single path. There are two distinct channels, and they are driven by different thematic ingredients.

Popularity (Reach)

Pillar Finding: ...is driven by themes of **Status & Safety**.

Perceived Quality

Pillar Finding: ...is driven by themes of **Care & Psychological Depth**.

Key Study Details: Dataset: 92 billionaire romance novels, 680,822 sentences.

Method: Neural topic modeling (BERTopic) + statistical analysis.

Defining the Two Channels of Success



Channel 1: REACH (Mass Appeal / Visibility)

What it is: How many people read and rate the book. It's a measure of market penetration and visibility.

How we measure it: ``log_rating_count`` from Goodreads.



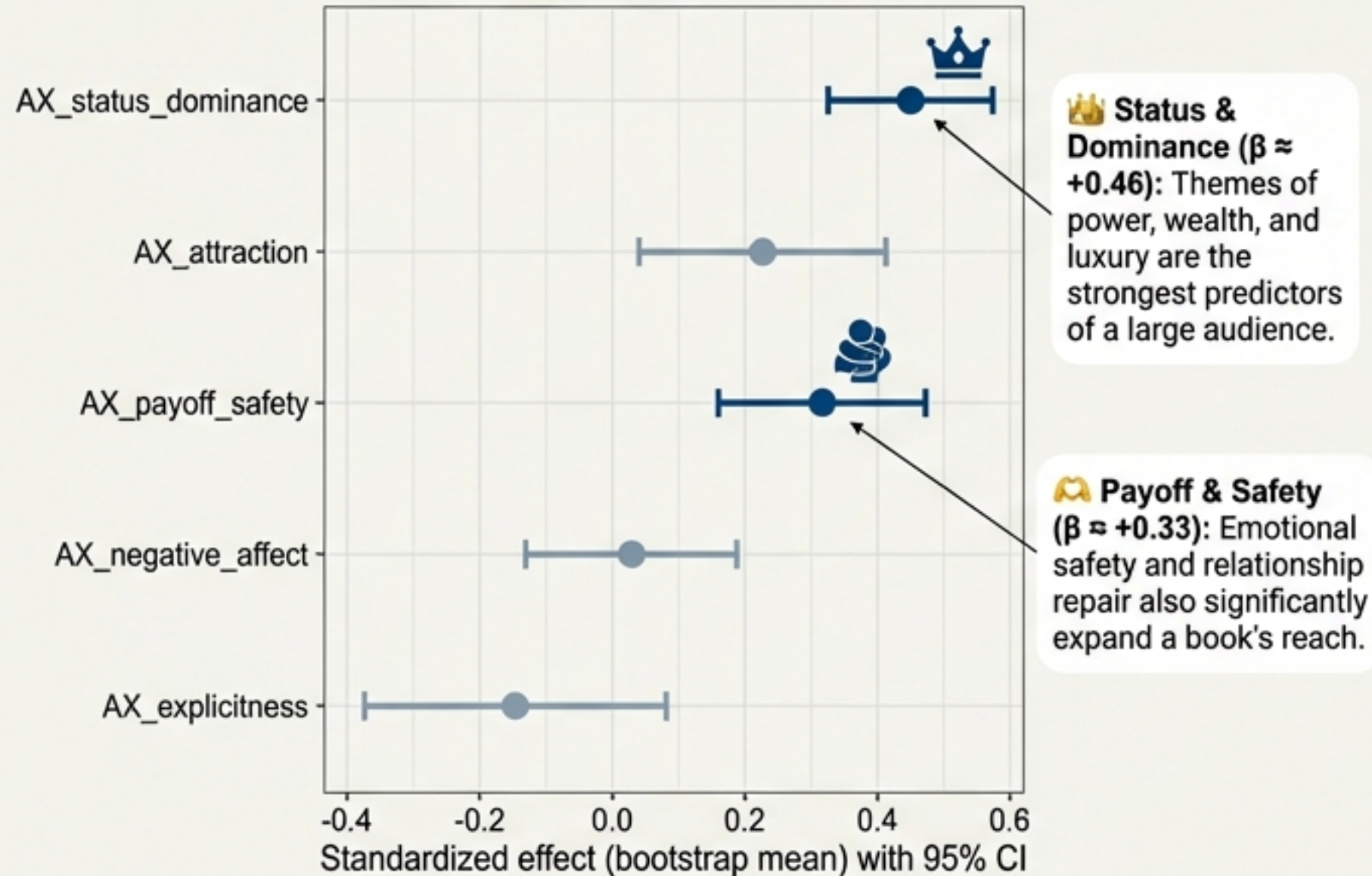
Channel 2: QUALITY (Perceived Reader Value)

What it is: How positively readers evaluate the book, independent of its popularity. It's a measure of reader appreciation.

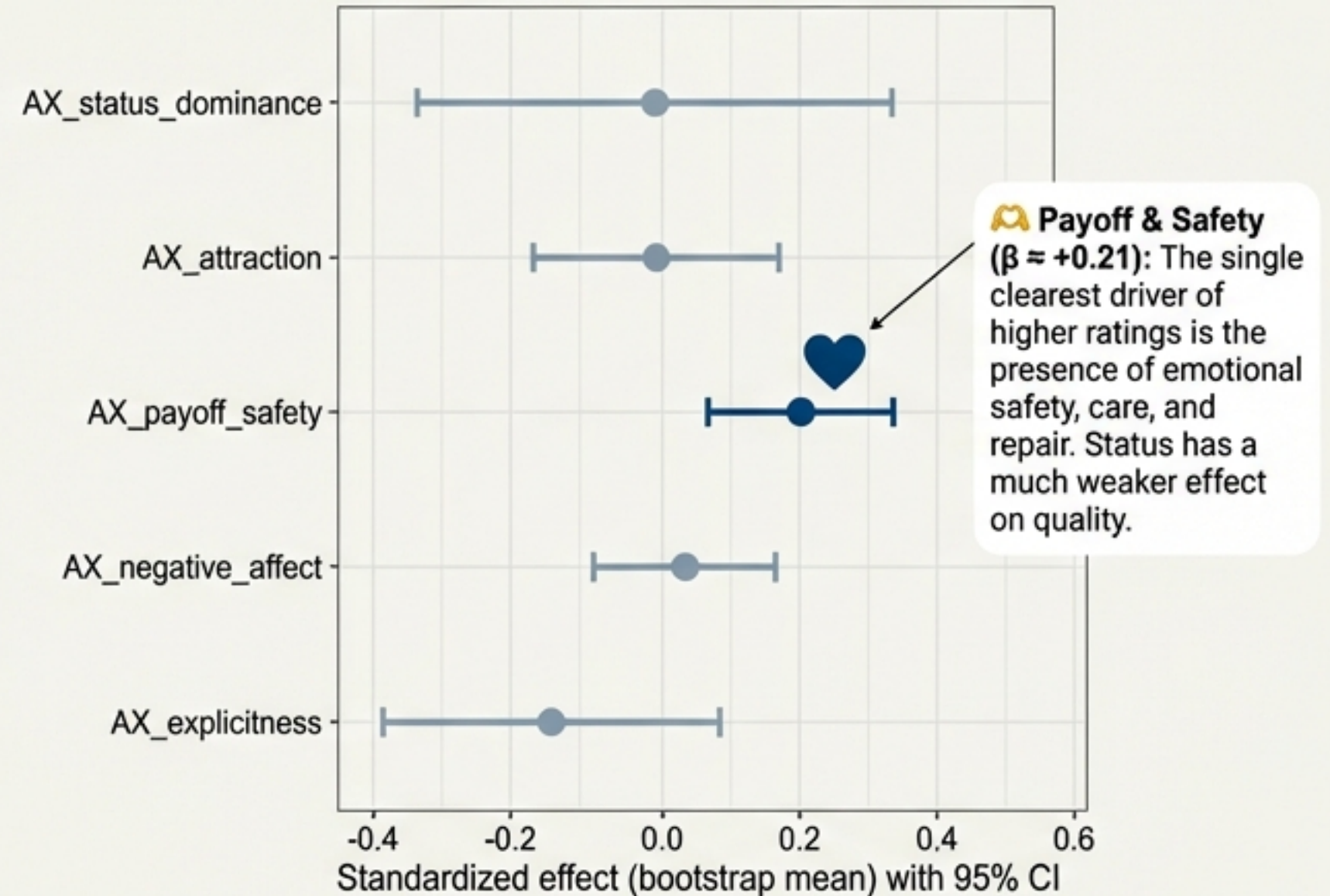
How we measure it: ``rating_mean`` from Goodreads, statistically controlling for the effects of reach.

The Thematic Drivers of Reach vs. Quality at a Glance

Macro-Themes Predicting Mass Appeal
(`log\_rating\_count`)



Macro-Themes Predicting Reader Appreciation
(`rating\_mean`, controlling for reach)



Notably, explicit sexual content is negatively associated with BOTH reach ($\beta \approx -0.27$) and quality ($\beta \approx -0.15$) in this corpus.

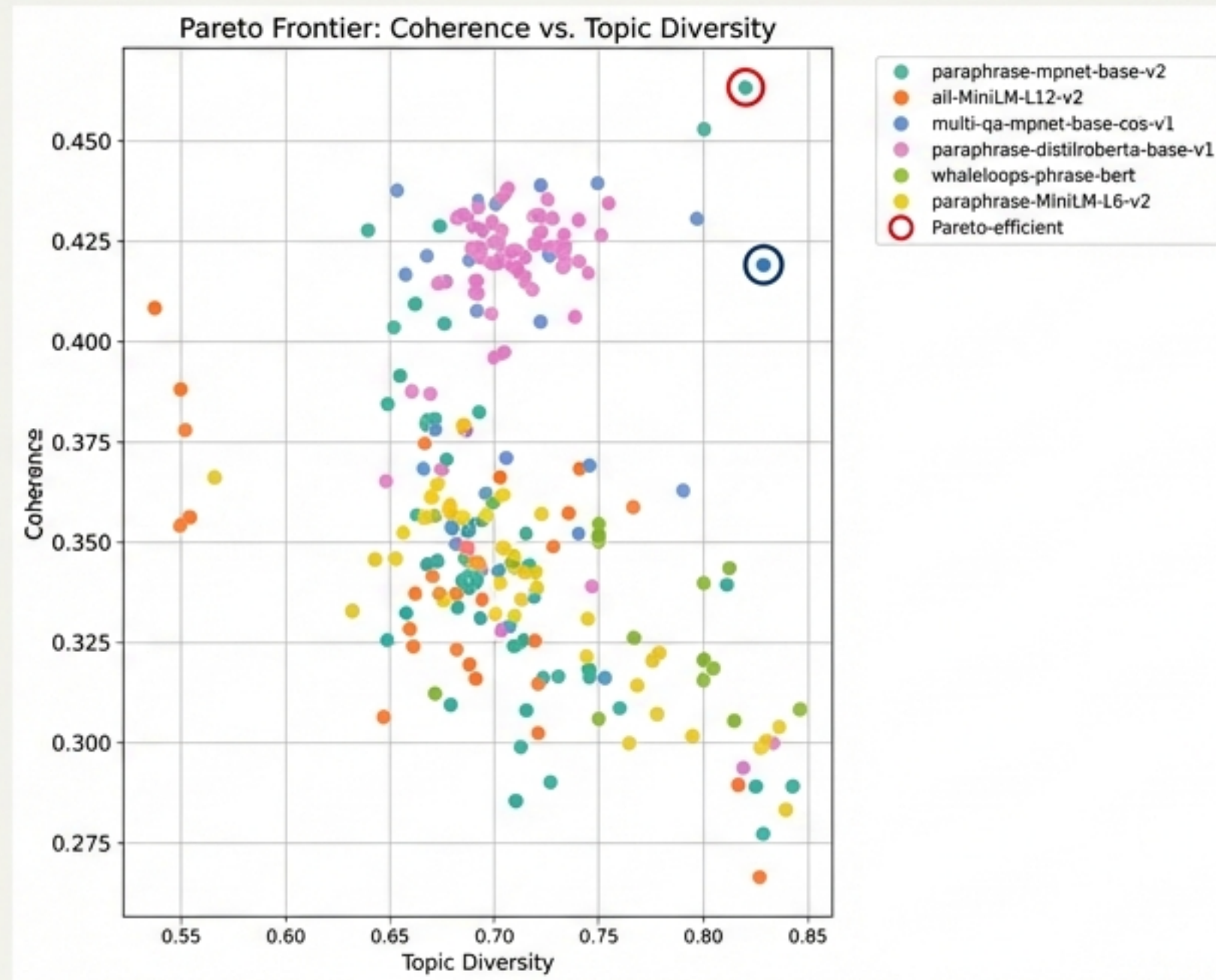
Building the Engine: How We Decode Fiction's DNA

The Challenge

Topic modeling involves a fundamental trade-trade-off. You can have highly coherent, interpretable themes, or you can have broad coverage (diversity). Optimizing for one often hurts the other.

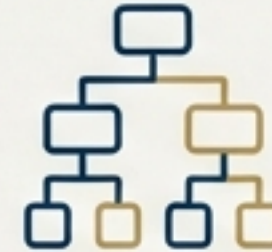
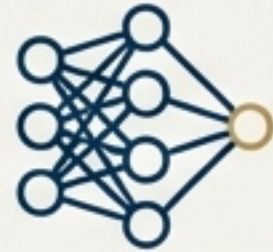
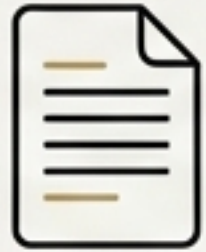
Our Solution

We used Pareto-Efficient Multi-Objective Optimization. Instead of guessing, we systematically tested over 300 different model configurations to find the set of models that offered the best possible trade-off between coherence and diversity.



Each point is a model. The points on the curve represent the optimal trade-offs. We chose our final model from this elite set, ensuring maximum analytical power.

From Raw Text to Thematic Fingerprints



Stage 1: Raw Text

680,822 sentences from 92 novels. We added 4,444 character names to our stopwords list to ensure topics focused on themes, not characters.

Stage 2: BERTopic

Our optimized neural topic model identifies clusters of sentences with similar meaning.

Output: 368 distinct "micro-scenes" or topics.

Stage 3: LLM Labeling

We used a Large Language Model (Mistral-Nemo-Instruct) to give each of the 368 topics a descriptive, descriptive, human-readable label.

Keywords like "car, parked, seat, kiss" became the labeled topic **"Parked Car Makeout."**

Stage 4: Theory-Aligned Taxonomy

The 368 micro-scenes were then mapped into a structured, **theory-driven taxonomy**.

Output: 8 main theme groups and 30+ specific categories.

"Parked Car Makeout" → Category: **Bonding, Everyday Intimacy**

Stage 5: Statistical Analysis

With themes quantified, we could **statistically test which ones** predicted **Reach** and **Quality**.

Higher-Rated Books Rebalance Their Thematic Budget

When comparing Top-tier vs. Trash-tier books, we see a subtle but consistent shift in where they focus their attention. Better books spend more of their “narrative space” on developing the couple’s relationship and their place in a wider social world.

➤ MORE in Top-Tier Books

- **Relationship Trajectory ($\delta \approx +0.37$):**
More focus on relationship development.
- **Social World Outside Couple ($\delta \approx +0.30$):**
More friends, family, and work context.
- **Embodied & Sensory Experience ($\delta \approx +0.29$):**
More grounded physical presence.

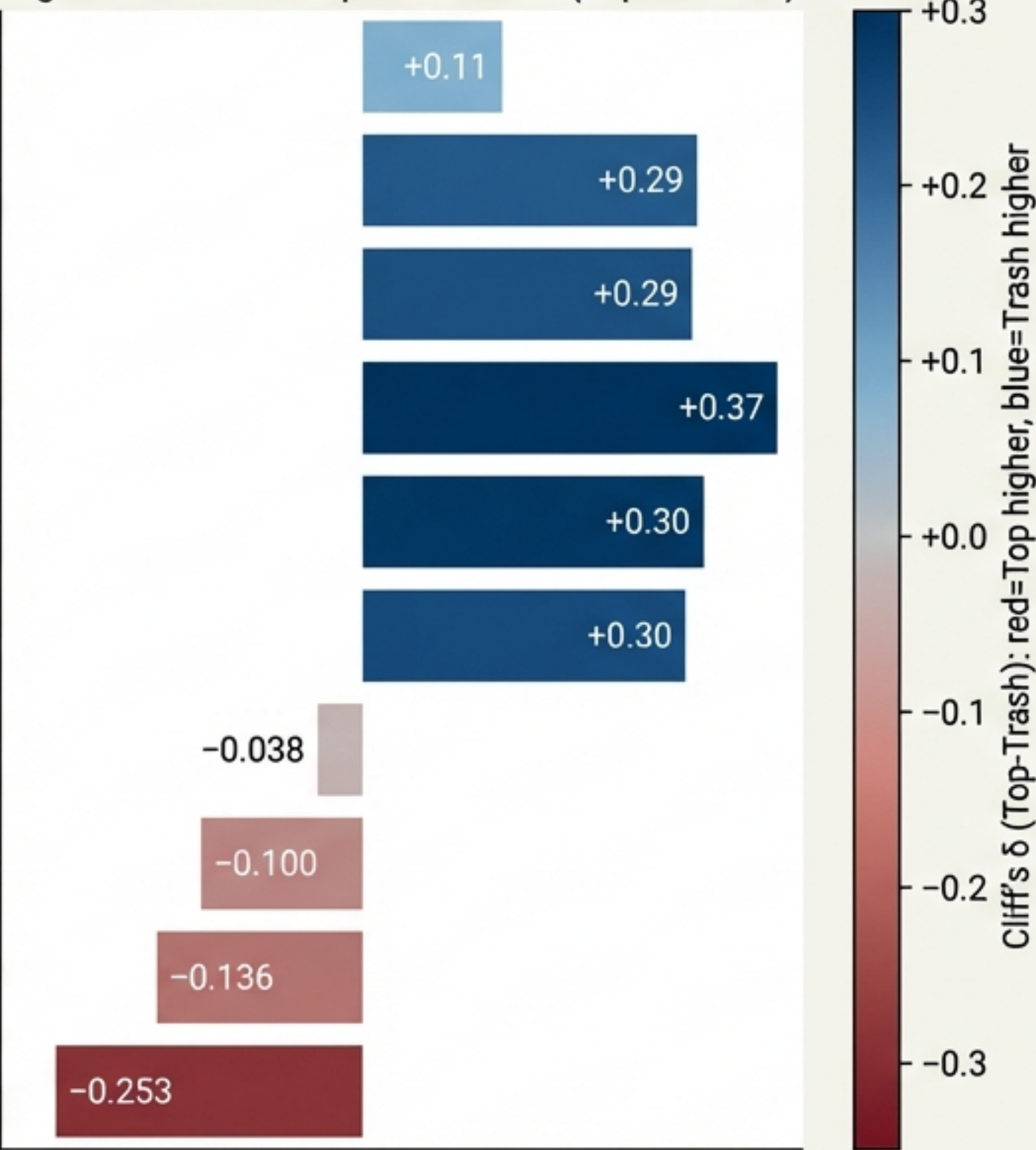
➤ LESS in Top-Tier Books

- **Sexuality, Attraction & Intimacy ($\delta \approx -0.25$):**
Less relative focus on pure sexual content.

Main Group

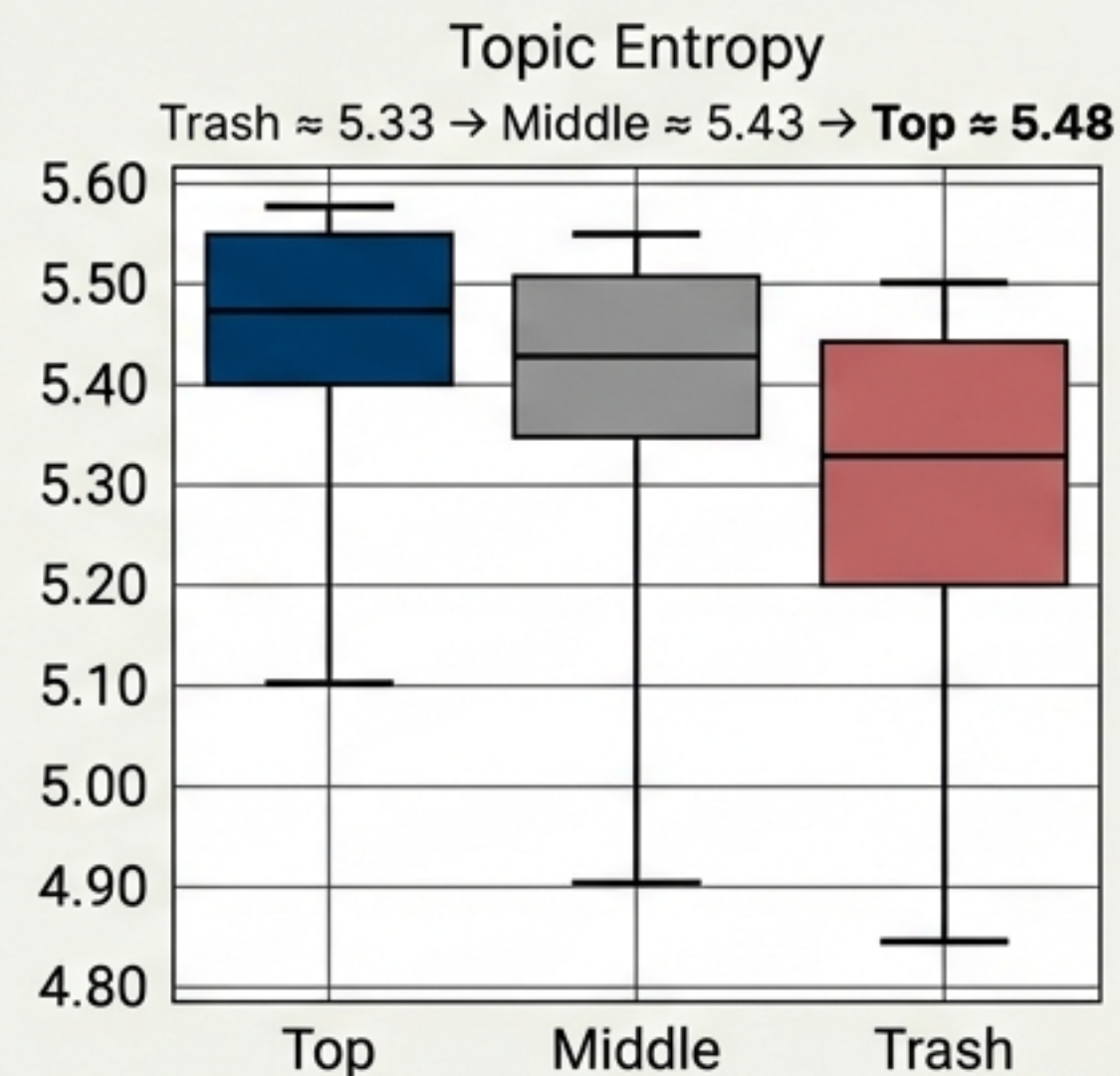
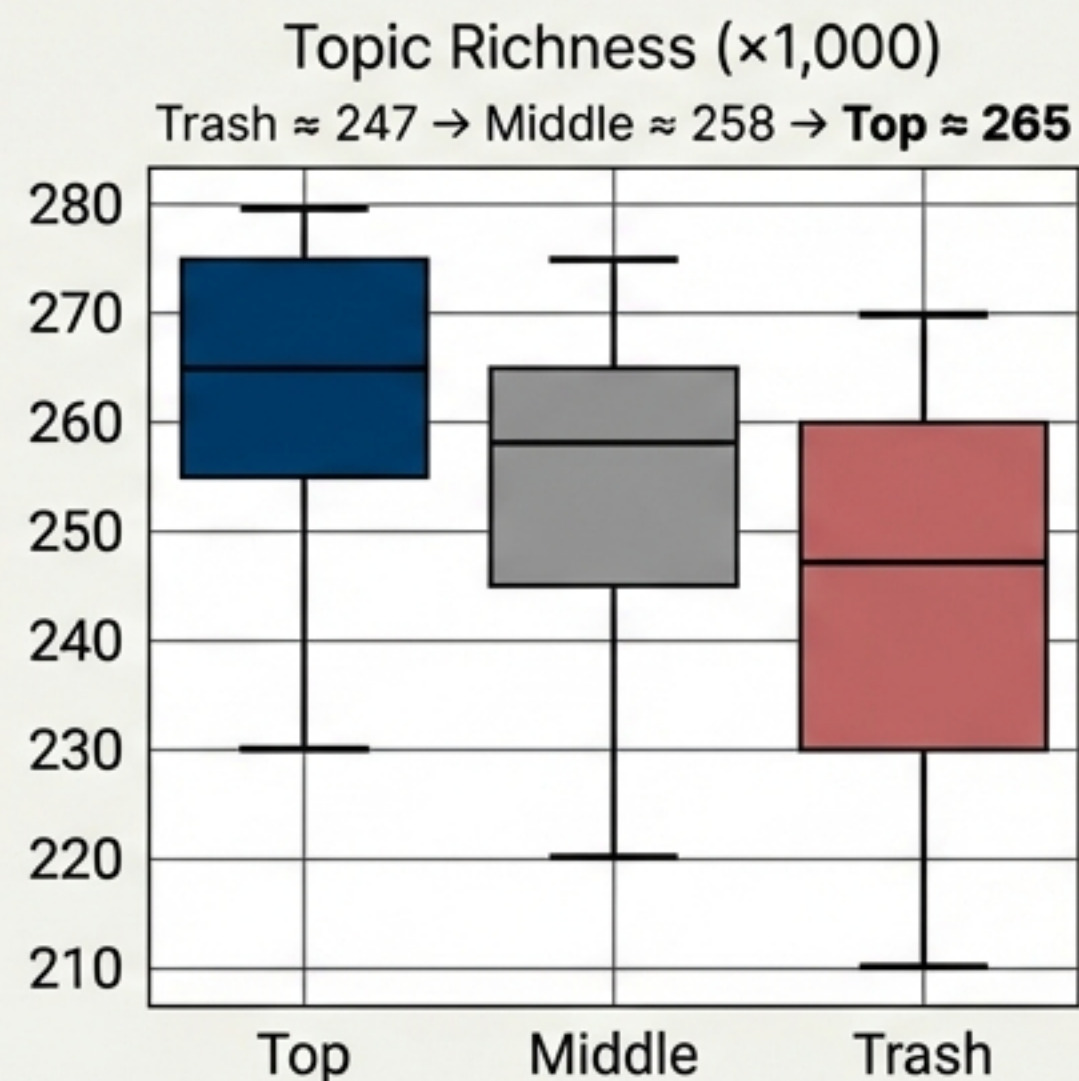
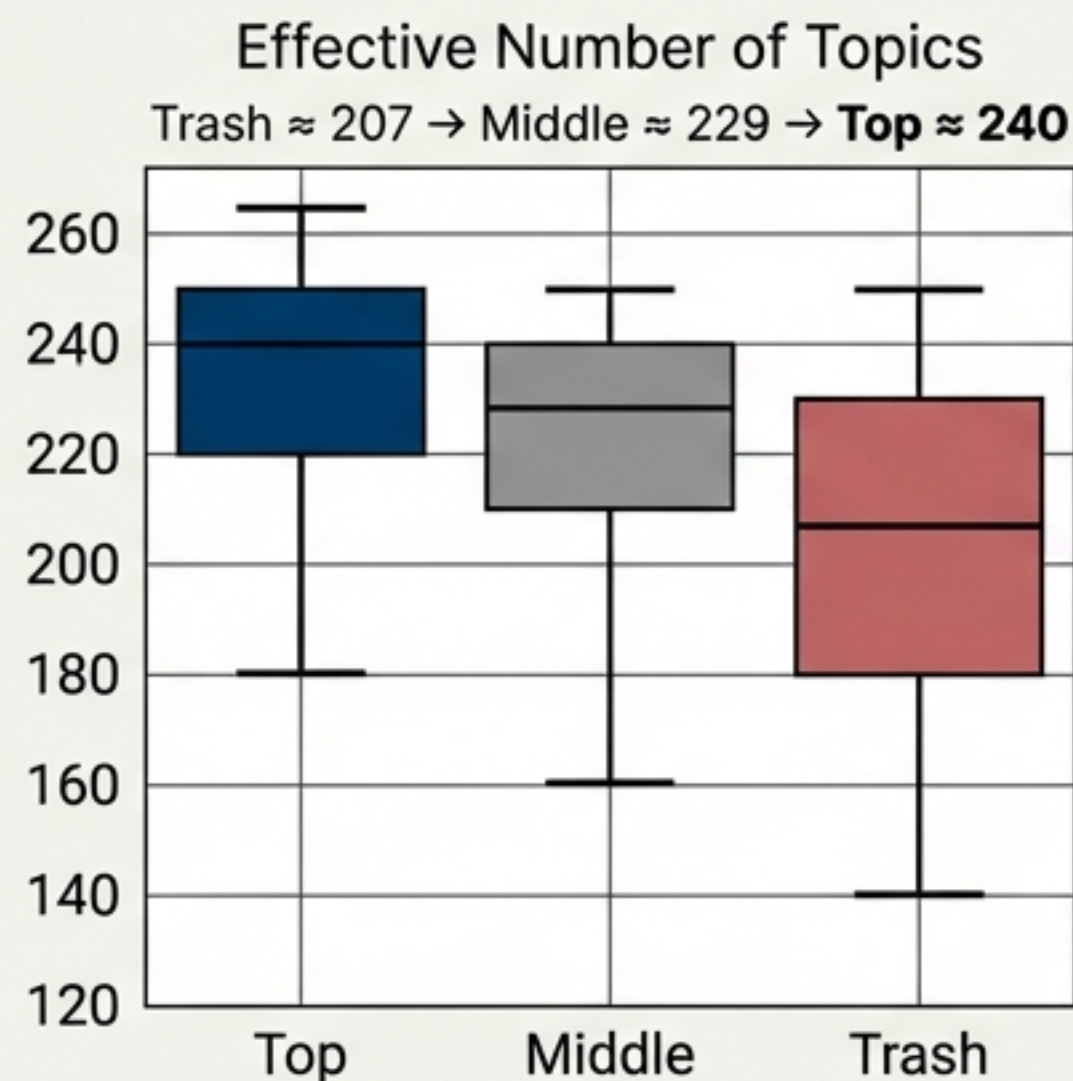
Conflict, Risk & Harm
Embodied & Sensory Experience
Emotions, Cognition & Inner Life
Relationship Trajectory (Main Couple)
Sexuality, Attraction & Intimacy
Social World Outside Couple
Spaces, Time, Activities & Objects
Work, Wealth, Status & Institutions
Emotions, Cognition & Inner Life
Sexuality, Attraction & Intimacy

Figure T2: Main Group Effect Sizes (Top vs Trash)



It's Not Just *Which* Ingredients, But *How Many*

Higher-tier books consistently show greater thematic diversity. They draw from a richer palette of topics, combining more unique thematic ingredients rather than over-concentrating on a few core motifs. This is a structural property of quality, not tied to any single trope.



Quality is in the Details: Psychological Depth vs. Procedural Filler

By analyzing the 368 specific 'micro-scenes,' we can pinpoint the exact moments that signal quality to readers.

TOP-TIER SCENES

Signal Psychological Credibility



'Married Couple's Affectionate Stares' ($\delta = +0.45$)

Small, non-verbal cues that build emotional intimacy.



'Frightened Admissions' ($\delta = +0.42$)

Scenes of vulnerability and emotional honesty.



'Feminist Identity Affirmation' ($\delta = +0.37$)

Moments of reflection on values and self-concept.

TRASH-TIER SCENES

Signal Procedural Filler or Low-Value Conflict



'Phone Notifications During Night'

Interruptions and transitional scenes that lack emotional depth.



'Dominatrix Session' ($\delta = -0.35$)

Specific types of explicit content that correlate negatively with ratings.

Timing is Everything: The Narrative Arc of a Great Story

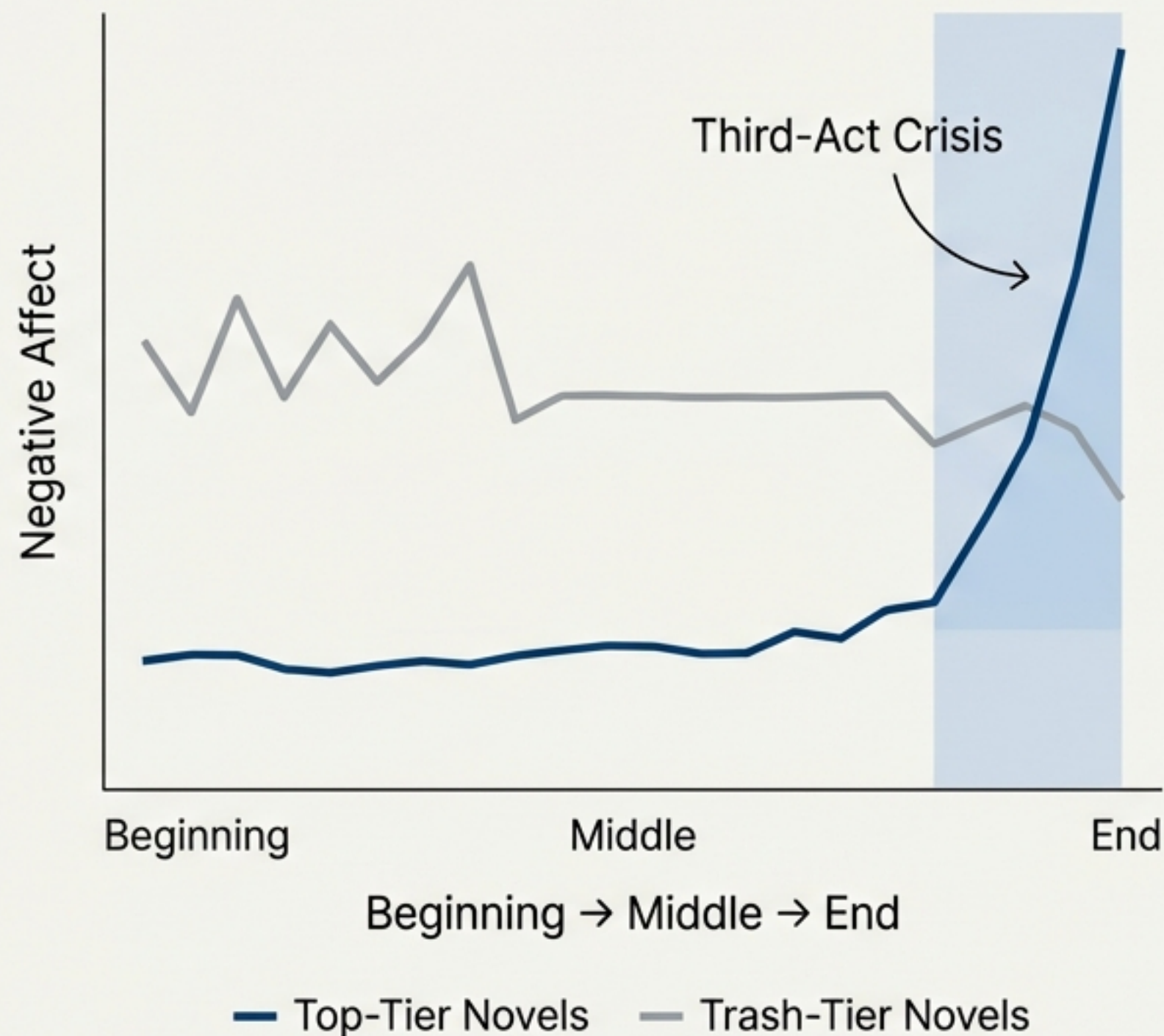
Higher-rated books don't avoid negative emotions; they deploy them strategically.

We found that better books have lower baseline negativity overall, but show a sharp increase in conflict and anxiety toward the end. We found that better books have lower baseline negativity overall, but show a sharp increase in conflict and anxiety toward the end.

This pattern maps directly to the classic "third-act crisis." A well-timed crisis makes the final happy ending feel more satisfying and earned.

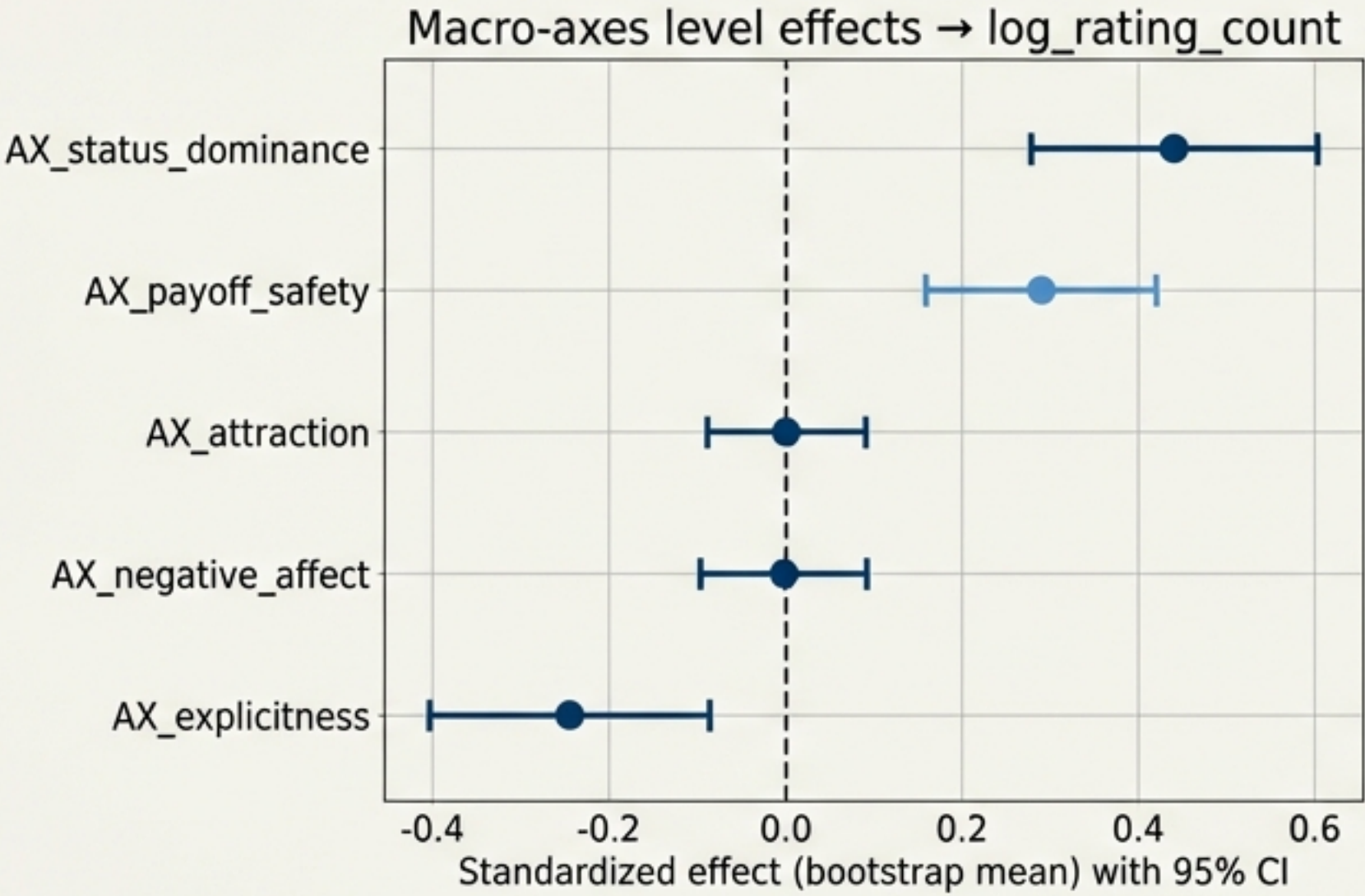
Predicting Higher Ratings:

- Increase in Anger/Frustration (end vs. begin): $\beta \approx +0.24$
- Increase in Anxiety/Worry (end vs. begin): $\beta \approx +0.19$



The Formula for REACH, Deconstructed

The specific thematic indices that most strongly predict a novel's mass appeal and visibility.



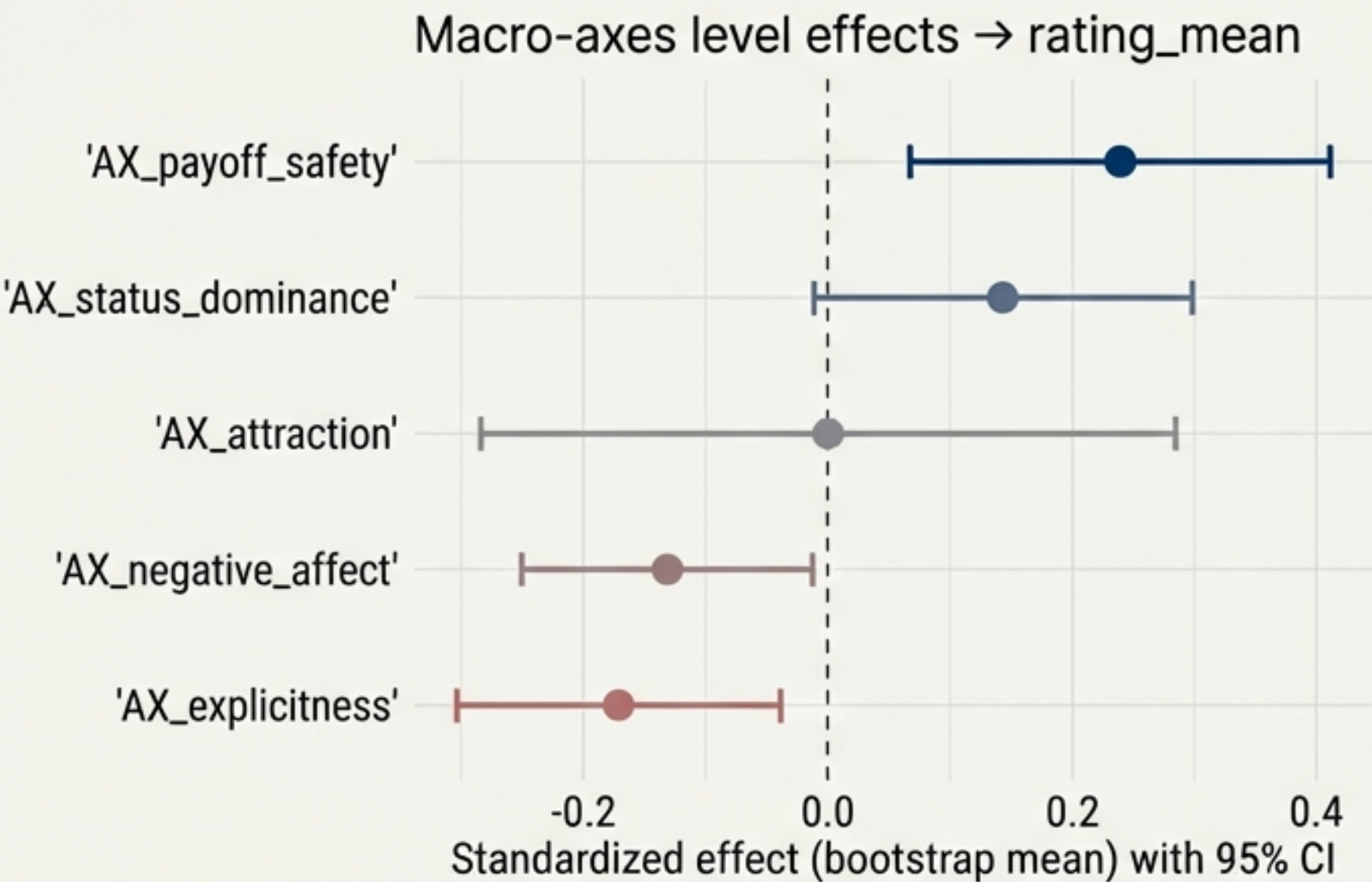
Top Predictors

Predictor	Beta Coefficient (β)	Confidence Interval	P($\beta>0$)
Macro-Axis: Status & Dominance	+0.46	[+0.30, +0.62]	1.00
Power, Wealth, Luxury	+0.37	[+0.37, +0.39]	1.00
Alpha Guarding	+0.44	[+0.44, +0.43]	1.00
Macro-Axis: Payoff & Safety	+0.33	[+0.18, +0.48]	P=0.999
Emotional Safety	+0.32	[+0.32, +0.56]	P=0.999
Relationship Repair	+0.23	[+0.23, +0.23]	P=0.999

The classic ‘billionaire romance package’—combining status, protectiveness, and the promise of a secure relationship—is a powerful formula for attracting a large readership.

The Formula for QUALITY, Deconstructed

The thematic indices that predict higher reader ratings, even after controlling for a book's popularity.



Top Predictors		
Predictor	Beta Coefficient (β)	Partial Correlation
Macro-Axis: Payoff & Safety	+0.21	
Protective Caretaking	+0.22	+0.245
Emotional Safety	+0.15	+0.152
Explicit Eroticism	-0.15	-0.172
Negative Affect	-0.13	-0.121

Quality is not about status or luxury. It is most strongly predicted by themes of care, protection, and emotional safety. These elements create a narrative that readers find genuinely rewarding.

The Two-Channel Success Model for Romance Novels



While ``Payoff & Safety`` themes contribute to both channels, REACH is primarily about **Status**, while QUALITY is primarily about **Care**.

The Blueprint: Actionable Insights for Stakeholders



For Authors

- To earn higher ratings, focus on **psychological credibility**. Use scenes of vulnerability ("Frightened Admissions"), reflections on values, and small, embodied intimacy cues ("Affectionate Stares").
- Master **narrative pacing**. A strategically placed late-story crisis can make the happy ending more potent.
- A richer, more diverse thematic palette is a hallmark of quality.



For Publishers

- Market **reach** is reliably driven by the "billionaire-romance package": status, luxury, and protective alpha-male behavior, paired with emotional safety.
- Recognize that the themes that drive mass-market sales are not identical to those that drive critical reader acclaim. Segment marketing and editorial strategies accordingly.



For Researchers

- The "Two-Channel" model provides a more nuanced framework for studying commercial fiction success.
- This study offers a replicable, mixed-methods pipeline for computational literary analysis, from model selection to theory-aligned interpretation.

Under the Hood: Our Commitment to Rigor

This pilot analysis was designed to build and calibrate a new measurement instrument for narrative themes. Our effect-size focused approach prioritizes identifying stable signals over chasing statistical significance.

Current Limitations

- **Sample Size:** This is a pilot study with N=92 books. Findings require validation in a larger corpus.
- **Author Confounding:** Some themes may be specific to a few high-output authors. Future work will control for author-level effects.
- **Thematic Indices:** Our macro-axes are robust, but still a simplification of complex narrative content.

Future Work

- **Large-N Study:** Expanding the analysis to the full corpus of 105+ books.
- **Refined Modeling:** Including author fixed effects and testing for interaction effects (e.g., does Luxury only work when paired with Care?).
- **Finer Time-Course Analysis:** Moving beyond thirds to analyze the precise trajectory of themes throughout the narrative.

Appendix & Further Reading

The data, code, and detailed reports for this analysis are available for review.

Repository

- ⊖ Code & Data: [\[Insert Full Project URL Here\]](#)
- 📄 Detailed Reports: [reports/01_stage_reports/tagel0_correlation_analysiss/](#)
- ⊖ Documentation: See [README.md](#) and [SCIENTIFIC_README.md](#)

Key Methodological Tools

- ⊖ Topic Modeling: BERTopic (Grootendorst, 2022)
- ⊖ Hyperparameter Optimization: OCTIS (Terragni et al., 2021)
- ⊖ Automated Labeling: Mistral-Nemo-Instruct

Theoretical Foundations

- ⊖ Radway, J. (1984). *Reading the Romance: Women, Patriarchy, and Popular Literature*.
- ⊖ Propp, V. (1928). *Morphology of the Folktale*.